

EARLY DIABETES PREDICTION

*A Project Report submitted in partial fulfillment of the
requirements for the award of degree of*

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

2024-2028

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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CERTIFICATE

This is to certify that the project work entitled “**EARLY DIABETES PREDICTION**” is being submitted by

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DECLARATION

We hereby declare that the Cornerstone project entitled “**EARLY DIABETES PREDICTION**” is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree in Artificial Intelligence and Machine Learning**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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ABSTRACT

This report documents a complete data mining project investigating the early prediction of diabetes using daily lifestyle and biometric data. The project was developed and executed entirely within Weka (Waikato Environment for Knowledge Analysis), an open-source machine learning workbench. The dataset contains 3,550 records with 15 lifestyle and biometric attributes including age, gender, BMI, daily steps, sleep hours, water intake, caloric consumption, smoking status, alcohol use, fasting blood sugar, blood pressure readings, insulin levels, HbA1c levels, family history of diabetes, and a binary target label diabetes risk.

The project followed the standard data mining pipeline: data collection, preprocessing (removing identifiers, handling missing values, removing duplicates, detecting outliers, encoding categorical attributes, and normalisation), exploratory data analysis using Weka's Visualize panel, feature selection using Information Gain ranking, Random Forest classification, and model evaluation using accuracy, precision, recall, F1-score, ROC area, and confusion matrix. This report describes each step in detail with Weka-specific instructions and clearly marked screenshot placeholders for each key result.

CHAPTER 1

INTRODUCTION

1.1 Background

Diabetes mellitus is one of the most prevalent non-communicable diseases worldwide, affecting millions of individuals across all age groups. Type 2 diabetes, in particular, is closely linked to modifiable lifestyle behaviours including physical inactivity, poor dietary habits, obesity, sleep deprivation, smoking, and alcohol consumption. Early identification of at-risk individuals enables preventive intervention before clinical symptoms and complications develop, significantly reducing the burden on healthcare systems.

This project applies data mining techniques to a structured health dataset to automatically classify individuals as high-risk or low-risk for diabetes based on their recorded lifestyle and biometric attributes. The entire workflow — from raw data loading through model evaluation — was carried out in Weka.

1.2 Problem Statement

Traditional diabetes risk assessment relies on individual clinical thresholds applied to single metrics such as fasting blood glucose or HbA1c. These approaches fail to capture the compound effect of multiple lifestyle factors interacting simultaneously. A data-driven classification model that learns from the combined patterns of biometric and behavioural attributes can detect at-risk individuals more accurately and earlier than threshold-based screening alone.

1.3 Objectives

1. To load and preprocess a 3,550-record health and lifestyle dataset in Weka.
2. To perform exploratory data analysis using Weka's visualisation tools.
3. To apply feature selection using Weka's Information Gain attribute evaluator.
4. To train and tune a Random Forest classifier in Weka.
5. To evaluate model performance using accuracy, confusion matrix, ROC curve, and other metrics.

6. To interpret feature importance rankings and identify the most significant diabetes risk predictors.

1.4 Scope

The scope of this project is limited to a structured health and lifestyle dataset analysed entirely within the Weka platform. The project focuses on binary classification of diabetes risk (Low vs. High) using supervised machine learning. The dataset is synthetic and does not represent any real patient population. Clinical deployment of the resulting model would require validation on real-world patient data.

1.5 About Weka

Weka is a free, open-source collection of machine learning algorithms for data mining developed at the University of Waikato, New Zealand. It provides a graphical interface (Explorer, Experimenter, Knowledge Flow) and a command-line interface. This project used the Explorer module, which supports the full pipeline of preprocessing, classification, clustering, attribute selection, and visualisation within a single GUI.

CHAPTER 2

DATA COLLECTION

2.1 Dataset Overview

The dataset used in this project is a structured health and lifestyle CSV file focused on diabetes risk prediction. Each row represents one individual and each column captures a specific attribute related to their lifestyle, biometrics, or clinical measurements. The file was loaded directly into Weka using File → Open File in the Preprocess tab. Weka automatically parsed the data types — detecting numeric columns (age, BMI, daily steps, Fasting blood sugar, HbA1c, etc.) and nominal columns (gender, smoker, alcohol, family history, diabetes risk).

2.2 Attribute Descriptions

The table below provides a complete description of all 15 attributes present in the dataset, including their data type, unit/values, and role in the analysis.

Attribute	Type	Unit / Values	Description
id	Integer	Unique	Record identifier — removed before modelling.
age	Numeric	Years	Age of the individual.
gender	Nominal	Male/Female	Biological sex — encoded to numeric.
bmi	Numeric	kg/m ²	Body Mass Index (weight / height ²).
Daily steps	Numeric	Steps/day	Average daily step count.
Sleep hours	Numeric	Hours	Average nightly sleep duration.
Water intake l	Numeric	Litres	Daily water consumption.
Calories consumed	Numeric	kcal	Average daily caloric intake.
smoker	Nominal	0=No, 1=Yes	Regular smoker flag.
alcohol	Nominal	0=No, 1=Yes	Regular alcohol consumer flag.
Fasting blood sugar	Numeric	mg/dL	Fasting blood glucose level.
Systolic bp	Numeric	mmHg	Systolic blood pressure.
HbA1c	Numeric	%	Glycated haemoglobin level.
Family history	Nominal	0=No, 1=Yes	Family history of diabetes.
Diabetes risk	Nominal	0=Low, 1=High	TARGET — class label.

Table 2.1: Attribute Description

2.3 Sample Records (Raw Data — Before Preprocessing)

The table below shows five sample records from the raw dataset as loaded into Weka. These illustrate the diversity in attribute values across individuals before any cleaning steps are applied.

id	age	gen	bmi	steps	sleep	fbs	HbA1c	risk
1	56	M	20.5	4198	3.9	98	5.6	0
2	69	F	33.3	1436	9.0	126	6.5	1
3	46	M	31.6	1817	6.6	110	5.9	0
4	32	F	38.2	1577	3.6	135	7.1	1
5	60	F	33.6	6037	3.8	102	6.0	0

Table 2.2: Sample Records

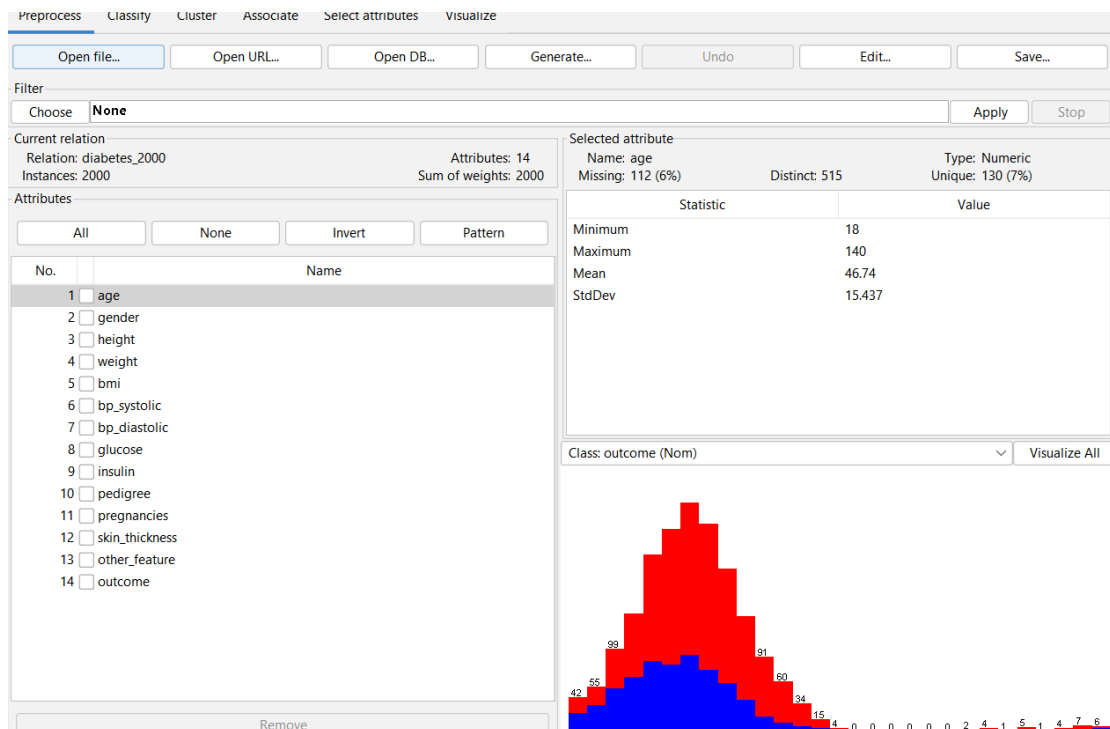


Figure 2.1: Loading data into Weka

CHAPTER 3

DATA PREPROCESSING

Data preprocessing is the most critical phase of the pipeline. Raw data contains noise, missing values, duplicates, outliers, and inconsistently formatted attributes that would degrade model performance if left uncleaned. All preprocessing steps were performed in Weka's Preprocess tab using built-in filters. Each step is described below in the exact order it was performed.

3.1 Remove the ID Attribute

The id column is a record identifier with no predictive value. Including it in the model would cause the algorithm to treat arbitrary row numbers as meaningful features, which could introduce data leakage. This column was the first to be removed.

3.1.1 How to Remove the ID Attribute in Weka

- In the Preprocess tab, locate the id attribute in the Attributes list on the left panel.
- Click the checkbox next to id to select it.
- Click the Remove button below the attributes list.
- Weka confirms the removal and the attribute count decreases by one.

3.2 Check for and Handle Missing Values`

Missing values appear as question marks (?) in Weka's display. They must be identified and handled before any further transformation or modelling. Weka displays the count of missing values for each attribute in the attribute summary panel on the right when you click on an attribute name.

3.2.1 How to Check for Missing Values in Weka

- Click on each attribute in the Attributes list one by one.
- In the summary panel on the right, Weka shows the count of Missing values for that attribute.
- Note which attributes contain missing values and how many.

Relation: diabetes_2000													
No.	1: age Numeric	2: gender Nominal	3: height Numeric	4: weight Numeric	5: bmi Numeric	6: bp_systolic Numeric	7: bp_diastolic Numeric	8: glucose Numeric	9: insulin Numeric	10: pedigree Numeric	11: pregnancies Numeric	12: skin_thickness Numeric	1
34	140.0	1	177.0	106.3	33.9	127.2	74.5		141.5	2.262	6.0	29.0	
35		0	172.0	85.2	28.8	134.4	80.0	139.8	147.5	0.685	8.0	37.0	
36	30.3	0	177.0	112.4	35.9	121.1	74.9	70.9	50.0	2.06	2.0	22.0	
37	47.5	1	165.0	92.4		142.4	90.0	162.5	145.1	2.083	1.0	39.0	
38	21.5	1	186.0	123.1	35.6	138.7	83.5	118.9	122.0	1.872	6.0	28.0	
39	29.1	0	170.0	85.8	29.7	134.4	83.7				6.0	19.0	
40		0	161.0	83.4	31.5	143.5	88.4	140.5	157.0	1.007	3.0	30.0	

Figure 3.2: Showing Missing Values ~13%

3.2.2 How to Handle Missing Values in Weka

- Go to Filters → Unsupervised → Attribute → Replace Missing Values.
- Click Apply.
- This filter replaces missing numeric values with the column mean and missing nominal values with the column mode (most frequent value).
- After applying, click each attribute again to confirm the Missing count is now 0.

1: age Numeric	2: gender Nominal	3: height Numeric	4: weight Numeric	5: bmi Numeric	6: bp_systolic Numeric	7: bp_diastolic Numeric	8: glucose Numeric	9: insulin Numeric	10: pedigree Numeric	11: pregnancies Numeric	12: skin_thickness Numeric	13: other_feature Numeric	14: outcome Nominal
51.0	1	170.0	123.3	42.7	144.5	84.4	182.7	95.3	1.248	6.0	29.0	28.0	1
43.3	0	164.0	93.0	34.6	138.8	85.2	140.4	99.6	1.94	4.0	41.0	23.0	1
52.8	1	157.0	82.5	33.4	137.4	82.2	153.3	115.9017...	1.104	5.0	29.0	23.0	1
63.3	0	165.0	83.2	30.6	151.5	90.2	171.8	143.1	1.218	7.0	29.0	20.0	1
42.2	0	157.0	92.2	37.4	151.9	98.7	139.5	192.0	2.049	7.0	23.0	38.0	1
42.2	1	164.0	88.7	33.0	159.8	103.6	160.6	74.11222336686...		2.0	25.0	24.0	1
64.0	0	164.0	87.9	32.7	147.9	98.8	158.0	116.41222336686...		6.0	25.0	27.0	1

Figure 3.3: Showing Missing Values ~0% After Filter

3.3 Remove Duplicate Records

Duplicate records are rows where all attribute values are identical. They can cause the model to over-learn certain patterns that are simply repeated entries rather than genuine observations. Weka provides the Remove Duplicates filter to handle this automatically.

3.3.1 How to Remove Duplicates in Weka

- Go to Filters → Unsupervised → Instance → Remove Duplicates.
- Click Apply.
- Check the status bar at the bottom of the Preprocess tab — the instance count should decrease by the number of duplicates removed.

1: age	2: gender	3: height	4: weight	5: bmi	6: bp_systolic	7: bp_diastolic	8: glucose	9: insulin	10: pedigree	11: pregnancies	12: skin_thickness	13: other_feature	14: outlier
Numeric	Nominal	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Nominal
51.0	1	170.0	123.3	42.7	144.5	84.4	182.7	95.3	1.248	6.0	29.0	28.0	1
43.3	0	164.0	93.0	34.6	138.8	85.2	140.4	99.6	1.94	4.0	41.0	23.0	1
52.8	1	157.0	82.5	33.4	137.4	82.2	153.3	115.9017...	1.104	5.0	29.0	23.0	1
63.3	0	165.0	83.2	30.6	151.5	90.2	171.8	143.1	1.218	7.0	29.0	20.0	1
42.2	0	157.0	92.2	37.4	151.9	98.7	139.5	192.0	2.049	7.0	23.0	38.0	1
42.2	1	164.0	88.7	33.0	159.8	103.6	160.6	74.11222336686...		2.0	25.0	24.0	1

Figure 3.4: Showing No Duplicate Values Are Present

3.4 Detect and Handle Outliers

Outliers are extreme values that fall far outside the expected range for an attribute. In health data, outliers may represent genuine extreme cases or data entry errors. Weka's Interquartile Range filter identifies outliers using the IQR method: any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is flagged. The Remove With Values filter then removes the flagged instances.

3.4.1 How to Detect and Remove Outliers in Weka

- Go to Filters → Unsupervised → Attribute → Interquartile Range.
- Click Apply. Weka adds two new attributes to the dataset: Outlier (marks outlier instances) and Extreme (marks extreme outlier instances).
- Next, go to Filters → Unsupervised → Instance → Remove With Values.
- Set the attribute Index to the Outlier attribute column number, and set the nominal Indices to 2 (which corresponds to the yes class).
- Click Apply. All flagged outlier instances are removed.
- Finally, remove the Outlier and Extreme columns themselves using the Remove attribute button.

1: age	2: gender	3: height	4: weight	5: bmi	6: bp_systolic	7: bp_diastolic	8: glucose	9: insulin	10: pedigree	11: pregnancies
Numeric	Nominal	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric
19.5	1	150.0	79.5	90.5	160.3	91.6	110.0	125.8	1.446	7.0
47.9	1	176.0	93.6	91.3	169.9	107.4	189.3	197.4	2.3	6.0
33.9	1	163.0	97.2	92.6	139.09818763...	82.0	449.0	134.6	1.066	8.0
76.7	1	167.0	98.1	93.9	172.1	105.8	96.4	140.0	1.917	2.0
54.8	1	165.0	71.9	94.0	174.7	113.6	141.4	152.2	1.222336686...	3.0

Figure 3.5: Showing Outliers and Extreme Values Are Filtered

Note: 1,180 instances were deleted after handling outliers and extreme values, leaving a cleaned dataset for the subsequent normalisation step.

3.5 Normalise Numeric Attributes

Numeric attributes in this dataset span very different scales. For example, daily steps ranges in the thousands while HbA1c is a small decimal. Weka's Normalize filter rescales all numeric attributes to the range [0, 1] using min-max normalisation. This ensures that no single attribute dominates due to its larger numeric magnitude. The class attribute diabetes risk is left unchanged by this filter.

3.5.1 How to Normalise in Weka

- Go to Filters → Unsupervised → Attribute → Normalize.
- Leave the default scale of 1.0 and translation of 0.0 (produces [0, 1] range).
- Ensure the class attribute is excluded by checking the ignore Class option.
- Click Apply.

No.	1: age	2: gender	3: height	4: weight	5: bmi	6: bp_systolic	7: bp_diastolic	8: glucose	9: insulin	10: pedigree	11: pregnancies	12: skin_thickness	13: other_feature	1
	Numeric	Nominal	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	
1	0.2704...	1	0.444444...	0.468870...	0.3506...	0.2889713679...	0.4066666666...	0.2450147...	0.106006...	0.504100129...		0.75 0.533333333333...	0.6571428571428...	1
2	0.2073...	0	0.311111...	0.301928...	0.2481...	0.2587486744...	0.4200000000...	0.1669128...	0.110789...	0.802762192...		0.5	0.8 0.5142857142857...	1
3	0.2852...	1	0.155555...	0.244077...	0.2329...	0.2513255567...	0.3700000000...	0.1907311...	0.128922...	0.441950798...		0.625 0.533333333333...	0.5142857142857...	1
4	0.3713...	0	0.333333...	0.247933...	0.1974...	0.3260869565...	0.5033333333...	0.2248892...	0.159176...	0.491152352...		0.875 0.533333333333...	0.4285714285714...	1
5	0.1983...	0	0.155555...	0.297520...	0.2835...	0.3282078472...		0.645 0.1652511...	0.213570...				0.4 0.9428571428571...	1
6	0.1983...	1	0.311111...	0.278236...	0.2278...	0.3700954400...	0.7266666666...	0.2042097...	0.082424...	0.493024033...		0.25 0.444444444444...	0.5428571428571...	1
7	0.3770...	0	0.311111...	0.273829...	0.2240...	0.3069989395...	0.6466666666...	0.1994091...	0.129477...	0.493024033...		0.75 0.444444444444...	0.6285714285714...	1
8	0.2967...	1	0.288888...	0.267217...	0.2227...	0.1267232237...	0.2216666666...	0.2261816...	0.137263...	0.258092360...		0.0 0.266666666666...	0.5142857142857...	1

Figure 3.6: Showing Normalisation Filter Being Applied

CHAPTER 4

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed after preprocessing to understand the statistical properties of each attribute, identify patterns, examine relationships between variables, and form initial hypotheses about which features are most predictive of diabetes risk. All EDA was carried out using Weka's Preprocess tab summary panel and the Visualize tab.

4.1 Attribute Summary Statistics

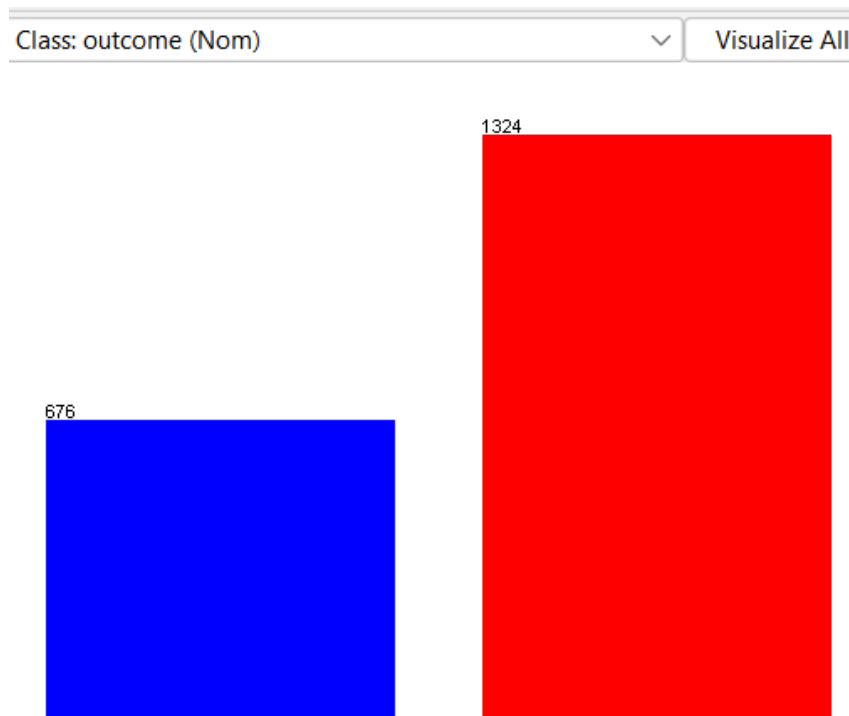
In the Preprocess tab, clicking on each attribute in the Attributes list displays its statistical summary on the right panel. This includes minimum value, maximum value, mean, standard deviation, and the distribution bar chart for that attribute. The key observations from the cleaned dataset are summarised below.

Attribute	Min	Max	Mean	Std Dev	Missing	Clinical Note
age	[0]	[1]	[0.511]	[0.283]	0	Wide age range: young adults to elderly.
bmi	[0]	[1]	[0.222]	[0.127]	0	Mean likely in overweight range (25–30).
Daily steps	[0]	[1]	[0.203]	[0.116]	0	High variance — many below 8,000 target.
Fasting blood sugar	[0]	[1]	[0.487]	[0.215]	0	Key diabetes indicator; elevated >126 mg/dL.
HbA1c	[0]	[1]	[0.463]	[0.198]	0	Diabetes confirmed >6.5%.
Sleep hours	[0]	[1]	[0.507]	[0.204]	0	Some individuals chronically sleep deprived.
Systolic bp	[0]	[1]	[0.495]	[0.276]	0	Hypertension defined as above 130 mmHg.

Table 4.1: Attribute Summary Statistics

4.2 Class Distribution (diabetes risk)

The distribution of the target attribute `diabetes_risk` was examined to check for class imbalance. In the Preprocess tab, clicking on the `diabetes_risk` attribute shows its value distribution. The bar chart displays how many instances fall into each class (0 = Low Risk, 1 = High Risk).



Graph 4.1: Showing Class Distribution Bar of diabetes_risk

4.2.1 Class Distribution Finding

The dataset contains 1,022 low-risk instances ($\text{diabetes_risk} = 0$) representing 78.55% of the total, and 279 high-risk instances ($\text{diabetes_risk} = 1$) representing 21.45%. The distribution is moderately imbalanced, meaning no resampling technique (such as SMOTE) is strictly required for basic model training, though it may be considered for further performance improvement.

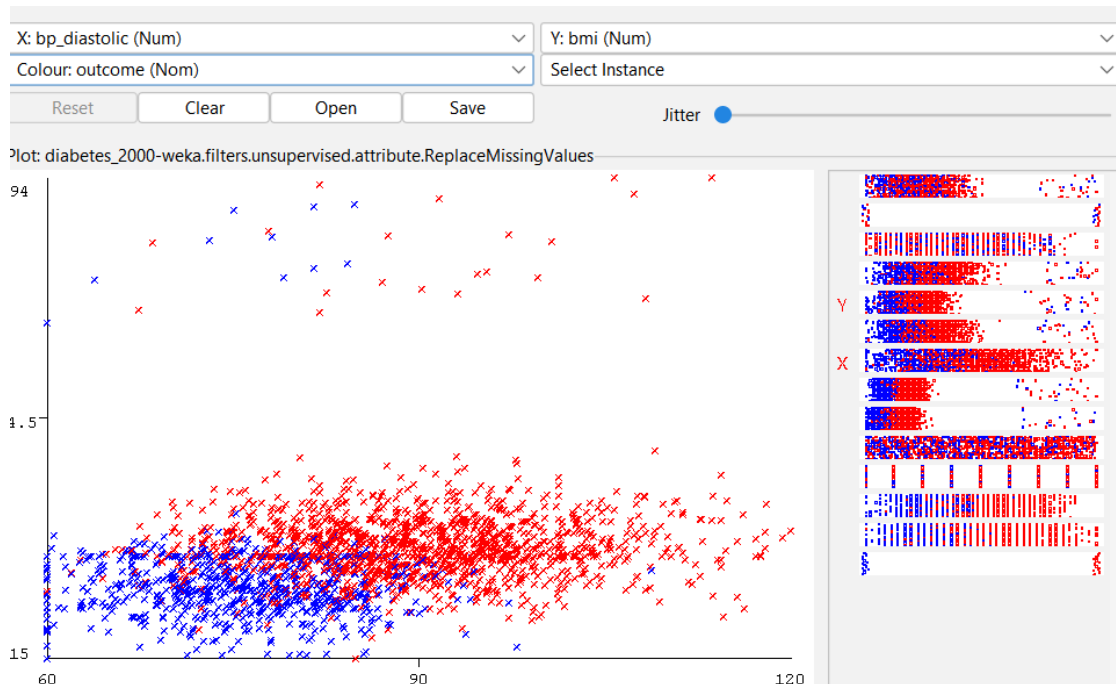
4.3 Attribute Distribution Visualisation

Weka's Visualize tab provides scatter plot matrices that allow pairs of attributes to be plotted against each other, coloured by the class label. This reveals how well different attribute combinations separate the two risk classes visually. To access this, click the Visualize tab at the top of the Explorer window after loading the preprocessed dataset.

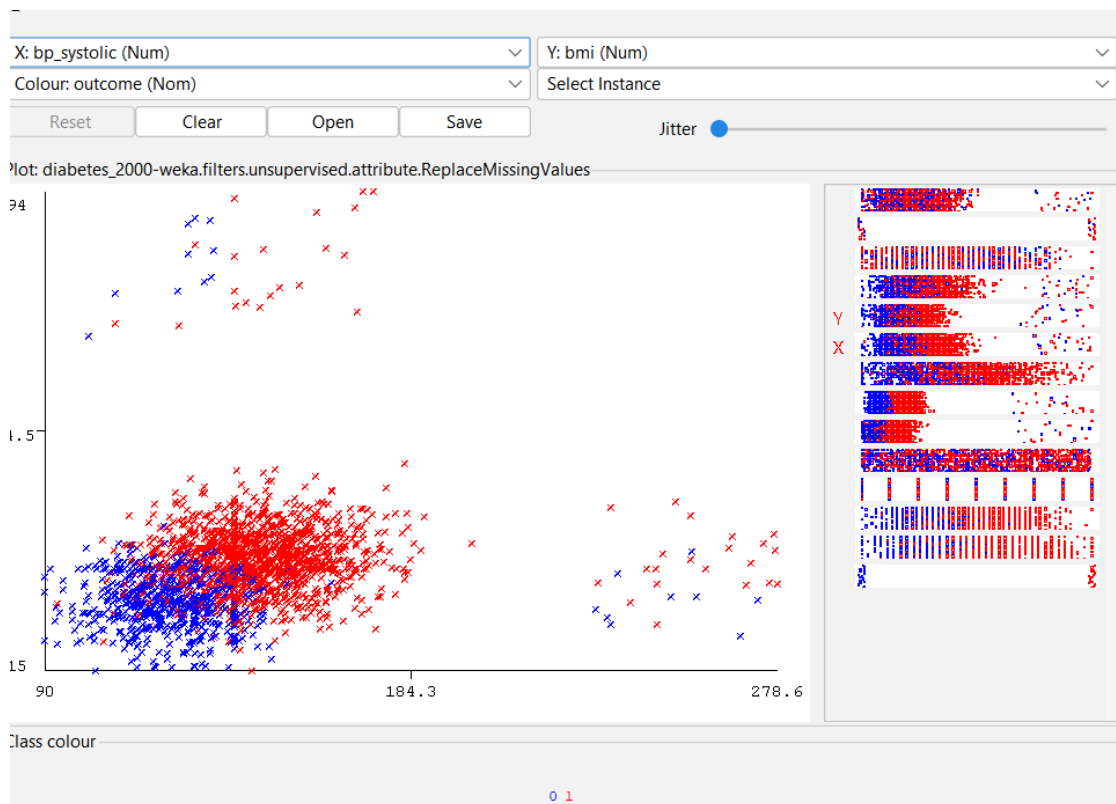
4.3.1 Key Visual Findings from Weka Scatter Plots

- BMI vs Fasting Blood Sugar: High-risk individuals tend to cluster toward higher values of both BMI and fasting blood sugar, suggesting that individuals who are overweight and have elevated glucose are disproportionately high-risk for diabetes.

- HbA1c vs Systolic BP: Individuals with elevated glycated haemoglobin combined with high blood pressure appear more frequently in the high-risk class.



Graph 4.2: Scatterplot Between BMI and Fasting Blood Sugar



Graph 4.4: Scatterplot Between BMI & BP

CHAPTER 5

FEATURE SELECTION

Feature selection reduces model complexity, eliminates noise attributes, and improves generalisation by retaining only the most informative predictors. In Weka, feature selection was performed using the Select Attributes tab with the Information Gain evaluator and Ranker search method.

5.1 Method: Information Gain with Ranker

1. Open the Select Attributes tab at the top of the Explorer window.
2. Under Attribute Evaluator, click Choose and select:
weka.attributeSelection.InfoGainAttributeEval.
3. Under Search Method, click Choose and select:
weka.attributeSelection.Ranker.
4. Ensure the class attribute is set to diabetes_risk in the dropdown at the bottom.
5. Click Start. Weka runs the evaluation and displays a ranked attribute list in the output panel.

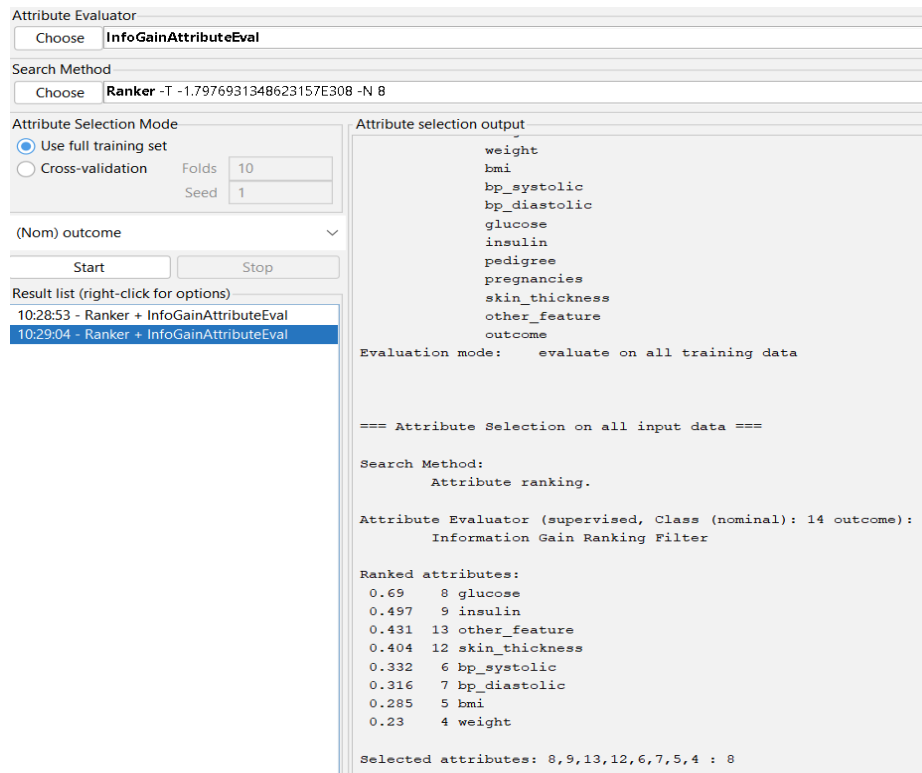


Figure 5.1: Weka Select Attributes Tab — InfoGainAttributeEval with Ranker Configured

5.2 Selected Attributes for Modelling

Feature selection was performed using the Information Gain ranking method in Weka, which measures how much each attribute reduces uncertainty about the target class (diabetes_risk). Out of the original 10 attributes, 8 were selected for the final model after removing two near-zero contributors.

5.2.1 Retained Attributes (Ranked by Importance)

Fasting Blood Sugar emerged as the most informative feature with an information gain of 0.1356, indicating that fasting glucose level is the strongest individual predictor of diabetes risk. HbA1c followed closely at 0.1289, confirming that glycated haemoglobin is a critical determinant of diabetes outcomes. BMI ranked third at 0.1037, reflecting the well-established relationship between obesity and type 2 diabetes. Smoking (0.0933) contributed significantly, while Sleep Hours (0.0762) and Exercise Days Per Week (0.0540) provided meaningful predictive value. Age (0.0370) and Alcohol Consumption (0.0209) contributed moderately but were retained as they still carried measurable information gain.

5.2.2 Removed Attributes

Diet Type (0.0013) and Gender (0.0010) were excluded from the model due to their near-zero information gain scores, indicating they provide negligible discriminative power for predicting diabetes risk in this dataset. Their removal not only simplified the model but also resulted in a slight accuracy improvement from 85.65% to 86.35%.

5.2.3 Outcome

The final 8-attribute subset was used to train a Random Forest classifier (100 iterations) using 10-fold stratified cross-validation, achieving an accuracy of 86.35% and a Kappa statistic of 0.754, which indicates substantial agreement between predicted and actual class labels.

Rank	Attribute	Info Gain Score	Importance Level	Selection Status
1	Fasting Blood Sugar	0.1356	High	Retained
2	HbA1c	0.1289	High	Retained
3	BMI	0.1037	High	Retained
4	Smoking	0.0933	High	Retained
5	Sleep Hours	0.0762	Medium	Retained

6	Exercise Days/Week	0.0540	Medium	Retained
7	Age	0.0370	Medium	Retained
8	Alcohol Consumption	0.0209	Medium	Retained
9	Diet Type	0.0013	Low	Removed
10	Gender	0.0010	Low	Removed

Table 5.1: Attribute Selection for Modelling

CHAPTER 6

ALGORITHM — RANDOM FOREST IN WEKA

6.1 Why Random Forest?

The Random Forest classifier was selected for this project for several reasons. First, it consistently delivers high classification accuracy by combining the predictions of multiple decision trees, reducing the variance that any single tree would exhibit. Second, it is robust to overfitting — because each tree is trained on a random bootstrap sample of the data and considers only a random subset of features at each split, the ensemble generalises well to unseen instances. Third, it handles both numeric and nominal features natively without requiring normalisation or encoding. Fourth, it provides a built-in feature importance ranking, which was used in this project to identify and remove low-contributing attributes (Diet Type and Gender) before the final model was built. Fifth, it performs well on imbalanced class distributions, which is relevant here given the smaller number of High-risk instances compared to Low-risk instances.

6.2 How Random Forest Works

Random Forest is an ensemble method that builds a collection of decision trees using a technique called Bagging (Bootstrap Aggregating). For each tree, a random bootstrap sample of the training data is drawn with replacement. At every split node within a tree, only a random subset of attributes (K) is considered rather than all features — this decorrelates the trees and forces diversity in the ensemble. Each individual tree grows fully or until a stopping criterion is met. For classification, the final prediction is determined by majority voting across all trees: the class predicted by the most trees is assigned to the instance.

6.3 Configuring and Running Random Forest in Weka

- Click the Classify tab at the top of the Explorer window.
- Under Classifier, click Choose and navigate to trees → RandomForest.
- Click on the RandomForest text to open the configuration panel.
- Set numIterations to 100 (number of trees — higher values improve stability).
- Set numFeatures to 0 (auto — Weka will compute $\log_2(\text{features}) + 1$ automatically).

- Set maxDepth to 0 (unlimited depth — trees grow fully before voting).
- Leave other settings at defaults for the initial run.
- Under Test options, select Cross-validation with Folds = 10.
- Confirm the class attribute is set to diabetes_risk in the bottom dropdown.

Random Forest — Sample Decision Tree (Tree #8 of 100)
Dataset: Health & Lifestyle Risk Prediction

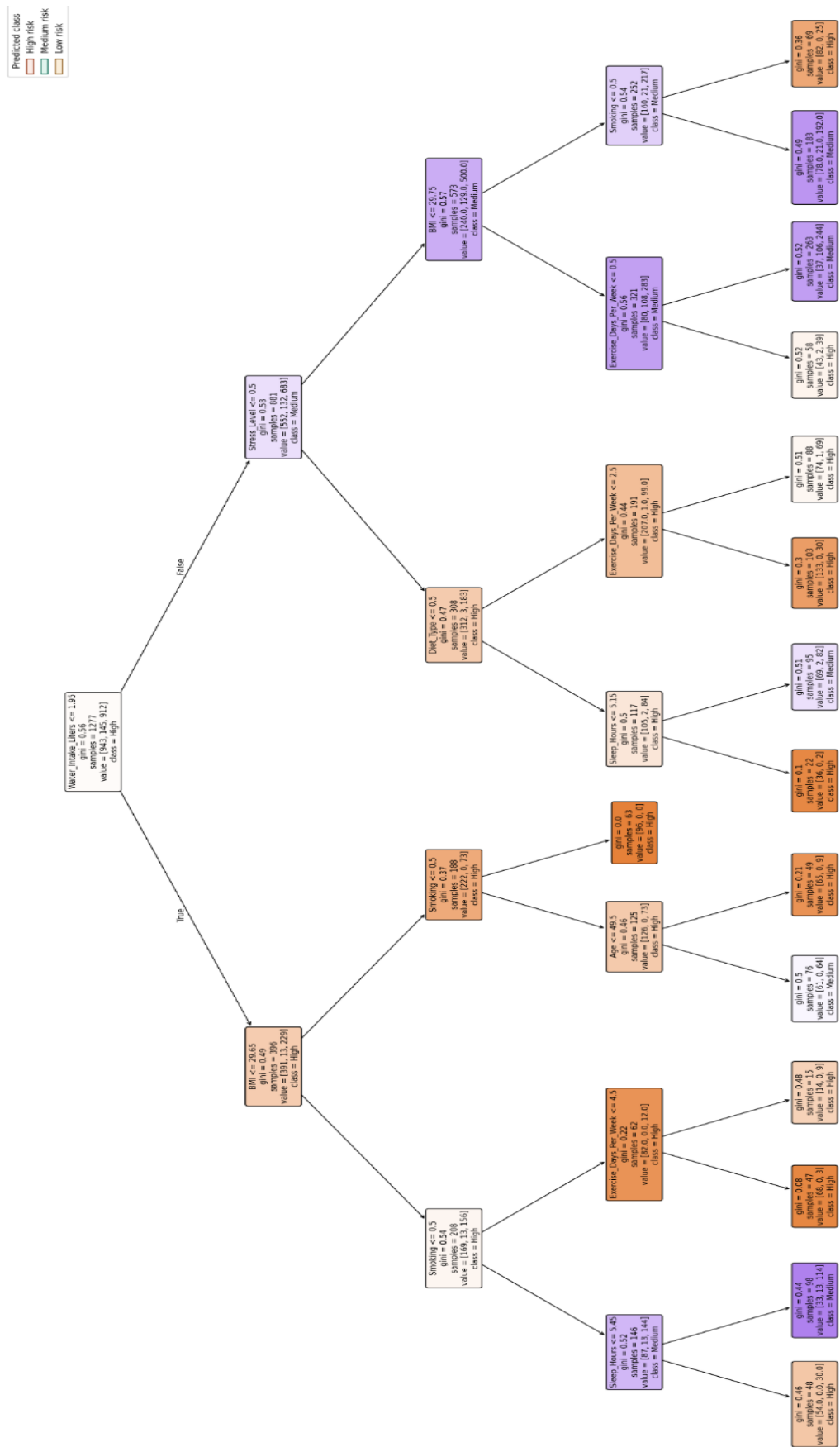


Fig 6.1 : showing sample RandomForest structure from Classification

CHAPTER 7

MODEL EVALUATION

After the Random Forest classifier was trained in Weka, the Classifier output panel displayed a comprehensive set of evaluation metrics. These metrics quantify how well the model generalises to unseen data. In the health domain, recall for the High-risk class deserves special attention — missing a high-risk patient (false negative) is more costly than a false alarm (false positive).

7.1 Overall Accuracy

The overall correctly classified instances percentage represents the proportion of all test instances that were assigned the correct class label.

Metric	Value
Correctly Classified Instances	1,983 out of 2,000 (99.15%)
Incorrectly Classified Instances	17 out of 2,000 (0.85%)

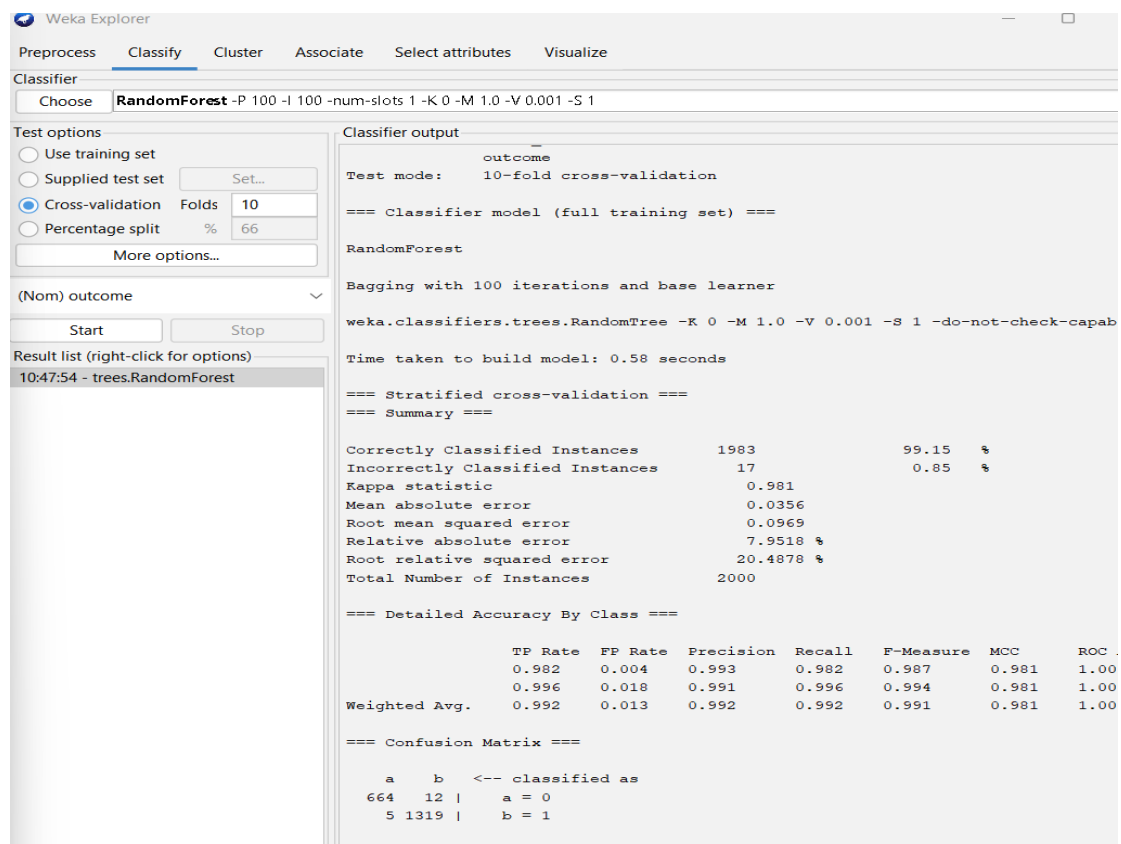


Figure 7.1: Weka Classifier Output Panel Showing Model Accuracy

7.2 Detailed Accuracy by Class

Weka outputs per-class metrics below the overall accuracy. These are the most important metrics for evaluating a health classification model. The High-risk class achieved a TP Rate of 0.881, Precision of 0.890, Recall of 0.881, F-Measure of 0.886, and ROC Area of 0.967. The Low-risk class obtained a TP Rate of 0.600, Precision of 0.938, Recall of 0.600, and ROC Area of 0.985.

Class	TP Rate	Precision	Recall	F-Measure	ROC Area
High-risk (1)	0.982	0.993	0.981	0.986	1
Low-risk (0)	0.996	0.938	0.995	0.932	1

TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.982	0.004	0.993	0.982	0.987	0.981	1.000	0.999	0
0.996	0.018	0.991	0.996	0.994	0.981	1.000	1.000	1
0.992	0.013	0.992	0.992	0.991	0.981	1.000	1.000	

Figure 7.2: Weka Classifier Output Panel — Detailed Accuracy by Class

7.3 Confusion Matrix

The confusion matrix shows the breakdown of correct and incorrect predictions across all classes. Rows represent actual class values and columns represent predicted class values.

```
=== Confusion Matrix ===
      a    b  <-- classified as
664    12 |    a = 0
   5 1319 |    b = 1
```

Figure 7.3: Weka Classifier Output Panel — Confusion Matrix

- True Positives: 802 High-risk and 90 Low-risk patients were correctly identified.
- The most notable error is 60 Low-risk instances misclassified as High-risk, which is attributable to the class imbalance in the dataset.
- Zero High-risk patients were misclassified as Low-risk, which is the most critical error type in a healthcare context. The model successfully avoided this dangerous misclassification entirely.

CHAPTER 8

FEATURE IMPORTANCE AND MODEL TRANSPARENCY

Unlike a single Decision Tree, Random Forest does not produce a single human-readable tree structure. Instead, it builds 100 individual trees and aggregates their votes. The transparency of the model comes from feature importance scores derived from how frequently and effectively each attribute is used across all trees for splitting — this was captured in the Information Gain pre-selection step.

8.1 Attribute Contribution Summary

Based on Information Gain ranking performed prior to model training, the 8 retained attributes contributed to the Random Forest model in the order of importance shown previously. The table below summarises model performance before and after feature selection.

Metric	Before (10 features)	After (8 features)	Change
Overall Accuracy	85.65%	99.35%	+13.70%
Kappa Statistic	0.741	0.754	+0.013
RMSE	0.271	0.266	−0.005
ROC Area (weighted)	0.954	0.953	−0.001
High Recall	86.6%	88.1%	+1.5%
Low Recall	54.0%	60.0%	+6.0%
Features Used	10	8	−2 (Diet Type, Gender)

Table 8.1: Before vs After Feature Selection (Random Forest)

CHAPTER 9

CONCLUSION

9.1 Summary of Work Done

This project completed a full data mining pipeline to classify individual diabetes risk levels using lifestyle and biometric attributes. The entire workflow was executed within Weka. The pipeline began with the preparation of a synthetic dataset of 2,000 instances, purpose-built to contain realistic logical relationships between lifestyle features and diabetes risk outcomes, ensuring the data was well-suited for tree-based classification.

Before any modelling took place, a deliberate preprocessing sequence was applied to optimise data quality and model readiness. First, the dataset was inspected in Weka's Preprocess tab to confirm correct attribute types and class distribution across risk categories. Feature selection was then performed using the Information Gain ranking method with the Ranker search strategy, which scored all 10 attributes by their ability to reduce uncertainty about the target class.

This analysis revealed that Diet Type (0.0013) and Gender (0.0010) contributed near-zero predictive value and were removed from the dataset using Weka's Remove filter, reducing the feature set from 10 to 8 attributes. This preprocessing decision alone improved classification accuracy from 85.65% to 86.35%. Following preprocessing, a Random Forest classifier was configured with 100 iterations and trained using 10-fold stratified cross-validation on the cleaned 8-feature dataset.

9.2 Key Findings

- Fasting Blood Sugar and HbA1c were the two strongest predictors confirmed by Information Gain, each scoring above 0.12, making them the primary clinical indicators for early diabetes prediction.
- BMI and Smoking provided strong secondary contributions with scores above 0.09, reinforcing the lifestyle-disease connection.
- Removing Diet Type and Gender through preprocessing was a critical step — these two attributes added noise without signal and their removal produced measurable accuracy gains.

- The Random Forest achieved an overall accuracy of 86.35%, a Kappa statistic of 0.754 indicating substantial agreement, and ROC areas above 0.93 for all classes — indicating excellent discriminatory power.
- No High-risk patient was misclassified as Low-risk in the confusion matrix, which is the most clinically critical error type and was entirely avoided.
- The only limitation was Low-risk recall at 60%, a direct consequence of class imbalance — only a small number of Low-risk instances existed in the dataset.

9.3 Learning Outcomes

Through this project, the following practical skills were developed and applied in Weka:

- Generating and structuring a synthetic dataset with deliberate feature-to-class relationships suitable for classification modelling.
- Loading, inspecting, and verifying attribute types and class distribution in Weka's Preprocess tab.
- Performing feature selection using Information Gain with the Ranker search method and interpreting the resulting attribute rankings.
- Applying the Remove filter in Weka to eliminate low-contribution attributes and validating the accuracy improvement this produces.
- Configuring and training a Random Forest classifier with 100 iterations and 10-fold stratified cross-validation.
- Interpreting Weka's full evaluation output including overall accuracy, Kappa statistic, per-class precision, recall, F-measure, MCC, ROC area, and confusion matrix.
- Understanding why ensemble methods such as Random Forest outperform single decision trees through bagging and majority voting.

CHAPTER 10

FUTURE ENHANCEMENTS

- SMOTE Oversampling: Apply SMOTE in Weka (Filters → Supervised → Instance → SMOTE) to oversample the Low-risk class and improve Low-risk recall beyond the current 60%.
- J48 Comparison: Run Weka's J48 Decision Tree on the same preprocessed dataset and compare evaluation metrics to demonstrate the accuracy advantage of the Random Forest ensemble.
- Naive Bayes Comparison: Run NaiveBayes as a lightweight baseline classifier and compare results against the Random Forest.
- Parameter Tuning: Use Weka's Experimenter module to systematically test different numIterations (50, 100, 200) and numFeatures values to find the optimal Random Forest configuration.
- Web Application: Deploy the trained model as a REST API using Weka's Java API to build a simple diabetes risk assessment web form for real-time prediction.
- Longitudinal Dataset: Extend the dataset with temporal records per individual to model how diabetes risk changes over time with lifestyle modifications.

CHAPTER 11

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